a



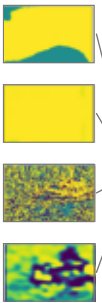
Input
448 × 448 × 3

CNN
ResNet-34 + UNet++

ViT
DINOv2 ViT-L/14

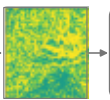
Noise
ELA + SRM

Color
Lab + local entropy



Evidence maps (1 ch)

Dual-attention fusion
channel + spatial
4 × 448 × 448



Fused map

128-d shared trunk

Authenticity
real / fake
BCE

Type
real / splice / copy-move / AI
weighted CE

Localization
pixel mask
BCE + Dice

$$\mathcal{L} = \mathcal{L}_{seg} + 0.5\,\mathcal{L}_{bin} + 0.5\,\mathcal{L}_{type}$$

Pixel-level supervision: BCE + Dice; type: class-weighted cross-entropy